

TYNDA, Russian Federation

WMO: 304990

Lat: 55.154N

Lon: 124.726E

Elev: 1734

StdP: 13.80

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-40.3	-36.6	-44.9	0.4	-39.8	-41.3	0.5	-36.1	12.3	-19.2	11.6	-19.6	4.2	250	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	84.2	65.2	80.6	63.8	77.2	62.5	68.4	78.4	66.6	75.6	64.8	73.3	2.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	99.8	71.6	63.5	93.5	70.2	61.7	87.5	68.8	33.8	78.3	32.3	75.6	30.9	73.5	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.3	9.7	8.5	DB	-45.1	89.9	4.7	2.8	-48.5	91.9	-51.2	93.5	-53.9	95.1	-57.3	97.1
			WB	-45.0	71.6	4.2	2.8	-48.0	73.6	-50.5	75.3	-52.9	76.9	-55.9	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.3	-17.9	-8.5	8.9	29.7	46.0	59.7	64.4	59.5	45.0	24.7	-3.1	-18.4
	DBStd	31.19	9.32	9.08	10.90	9.04	7.20	6.45	5.43	6.42	7.64	10.85	12.37	9.62
	HDD50	10495	2104	1639	1276	611	165	7	0	8	187	783	1594	2121
	HDD65	14941	2569	2059	1741	1060	589	179	75	190	601	1248	2044	2586
	CDD50	1129	0	0	0	1	41	300	448	303	36	0	0	0
	CDD65	99	0	0	0	0	0	21	58	20	0	0	0	0
	CDH74	1493	0	0	0	0	28	454	721	284	6	0	0	0
	CDH80	362	0	0	0	0	2	102	200	58	0	0	0	0
Wind	WSAvg	3.2	4.5	3.8	3.7	3.5	3.1	2.3	2.0	2.1	2.6	3.1	3.5	4.5
Precipitation	PrecAvg	23.9	0.3	0.3	0.4	1.1	2.3	4.4	5.4	4.9	2.6	1.3	0.7	0.4
	PrecMax	38.3	1.0	0.6	1.4	3.4	4.4	21.0	12.0	10.7	5.9	3.5	2.2	1.1
	PrecMin	16.5	0.0	0.0	0.0	0.1	0.8	0.4	0.6	0.6	0.7	0.2	0.2	0.0
	PrecStd	5.0	0.2	0.2	0.4	0.7	0.9	3.8	2.6	2.3	1.3	0.8	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	22.9	42.0	64.5	78.5	87.4	89.4	85.4	73.8	54.7	30.7	11.4
		MCWB	7.4	18.2	32.9	47.8	57.0	63.8	67.4	66.0	59.3	43.7	26.3	9.4
	2%	DB	4.7	16.4	36.3	56.3	71.4	82.8	85.5	80.9	67.3	48.5	24.4	5.8
		MCWB	3.2	13.5	29.1	42.7	52.6	62.5	67.2	64.7	55.8	40.0	22.4	4.4
	5%	DB	0.5	11.3	31.4	49.6	66.6	79.3	81.5	76.8	62.9	43.7	20.3	0.5
		MCWB	-0.7	9.0	25.6	38.7	50.4	61.4	65.3	63.6	53.0	36.7	18.2	-0.5
	10%	DB	-3.2	6.7	27.5	45.3	61.9	75.6	77.8	72.8	58.8	39.4	15.7	-3.9
		MCWB	-4.3	4.8	23.0	35.9	48.2	59.6	64.2	62.3	50.6	33.9	13.9	-4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	18.5	33.3	48.2	58.0	68.4	71.5	69.3	61.0	44.6	26.4	9.6
		MCDB	9.2	22.1	41.7	62.9	75.0	79.8	83.4	77.5	69.8	52.4	29.9	11.2
	2%	WB	3.2	13.3	29.2	42.9	54.7	65.5	69.3	67.2	57.7	40.9	22.3	4.5
		MCDB	4.7	16.2	35.7	55.6	68.1	76.9	80.1	76.0	64.5	48.1	24.4	5.8
	5%	WB	-0.7	8.9	26.1	39.2	51.7	63.4	67.5	65.3	54.8	37.2	18.3	-0.5
		MCDB	0.4	11.2	31.1	49.0	63.4	75.2	77.1	73.8	60.1	43.0	20.2	0.3
	10%	WB	-4.2	4.9	22.9	36.3	49.1	61.4	65.9	63.4	52.0	34.1	13.9	-4.7
		MCDB	-3.3	6.7	27.4	44.3	59.7	72.4	74.2	71.1	57.7	39.0	15.5	-4.0
Mean Daily Temperature Range	5% DB	MDBR	18.2	24.5	25.3	21.8	22.6	24.6	21.6	22.4	20.9	18.9	19.2	15.6
		MCDBR	18.9	27.1	27.5	28.9	33.3	32.9	30.8	30.5	28.7	23.6	20.0	16.2
		MCWBR	17.7	24.2	21.1	17.5	17.8	14.7	14.0	15.4	16.9	15.6	17.3	15.3
	5% WB	MCDBR	18.9	26.9	26.1	27.0	29.4	26.9	24.1	24.0	22.9	22.6	19.5	16.2
		MCWBR	17.7	24.0	20.1	16.7	16.4	13.6	12.1	12.9	14.5	15.2	17.0	15.3
Clear-Sky Solar Irradiance	taub	0.240	0.271	0.313	0.426	0.485	0.458	0.461	0.399	0.346	0.315	0.262	0.236	
	taud	2.119	2.104	2.044	1.863	1.926	2.124	2.157	2.327	2.416	2.292	2.121	2.053	
	Ebn at Noon	220	254	268	248	241	251	247	258	259	236	204	187	
	Edh at Noon	19	29	39	56	56	46	44	35	28	24	19	16	
All-Sky Solar Radiation	RadAvg	335	698	1272	1736	1607	1799	1580	1352	998	645	388	239	
	RadStd	14	23	46	101	113	150	160	143	90	45	24	9	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page